Avi Gupta

B-416, R&D Block, IIIT Delhi, India
www.iiitd.edu.in/~avig avig@iiitd.ac.in avigupta2798 avigupta

Summary

I am a fifth-year PhD student at IIIT Delhi, advised by Dr. Koteswar Rao Jerripothula and Dr. Tammam Tillo. My research interests include Computer Vision and Deep Learning. In particular, I work in Semantic Understanding of Large Multimodal Models, Continual Learning, and Camouflaged Scene Understanding. I plan to submit my thesis by December 2026 and graduate by July 2027.

Education

Indraprastha Institute of Information Technology, Delhi **Aug. 2022 – Present**
Doctoral Student (Ph.D.) in Computer Science *Delhi, Delhi*

Dr. Akhilesh Das Gupta Institute of Technology & Management, Delhi **Aug. 2017 – Jul. 2021**
Bachelor of Technology in Information Technology (8.56/10) *Delhi, Delhi*

Relevant Coursework

- Machine Learning
- Advanced Machine Learning
- Advances in Deep Learning
- Artificial Intelligence
- Computer Vision
- Digital Image Processing

Teaching Experience

- Meta-Learning
- Database Management System
- Introduction to Programming
- Digital Image Processing
- Data Structures and Algorithms
- Artificial Intelligence

Industry Experience

Dolby Laboratories **Jan 2026 – Apr 2026**
Research Intern *Bangalore, Karnataka*
Worked on efficient Class-Incremental Learning in Audio-Visual Multimodality to deploy on edge devices.

GE Aerospace **Aug 2025 – Dec 2025**
Research Intern *Bangalore, Karnataka*
1) Built an LLM-powered pipeline (via AWS Bedrock) to extract and structure unstructured aircraft engine data (via AWS S3), storing results in a SQL database for downstream processing.
2) Leveraged X-ray images of engine blades to detect and localize the patterns of scratches, dents, etc.

Tata Consultancy Services **Jul 2021 – Aug 2022**
Cloud Developer (SME) *Delhi, Delhi*
NFV OpenStack technology to deploy cloud services (Python, OpenStack, Shell Scripting, OpenSource Mano)

Digital Green **Jun 2019 – Aug 2019**
Backend Developer Intern *Delhi, Delhi*
Script Development for Automation of the Monitoring mailer for Data and Revenue reconciliation.

Research Projects

Relational context, from inter-object relationships **Sept 2026 – Present**
Ongoing
Relation-faithful customization. Given personalized subjects, generate scenes in which the specified relations hold.

Toward Genuine Forgetting: Selective Unlearning in Large Language Models **Aug 2026 – Present**
Ongoing
Working on understanding how to unlearn requested knowledge in LLMs in a continual setting.

Referring Camouflaged Object Discovery **Sept 2025 – Present**
Extension Ongoing for Journal *WACV 2025*
Working on extending the published work to incorporate videos and other scenarios.

Publications

Avi Gupta, Trasha Gupta, "Seeing the Unseen: Camouflaged Object Detection Beyond the Visible Spectrum", in *Fortieth Annual Conference on Neural Information Processing Systems Workshop (NeurIPSw), 2026*

Avi Gupta, Saurabh Yadav, Koteswar Rao Jerripothula, and Tammam Tillo, "SELECT: SELEctive Context Transfer in Class-Incremental Semantic Segmentation," in *Proceedings of British Machine Vision Conference (BMVC), 2026*

Avi Gupta, Nilotpal Sinha, Vishnu Raj, Sambuddha Saha, Pratik Joshi, Koteswar Rao Jerripothula, and Tammam Tillo. "Listen, Look, and Learn: Learning Without Forgetting through SAM-Audio", in *International Conference on Machine Learning Workshops (ICMLw), 2026.*

Saurabh Yadav[1], **Avi Gupta**[1] and Koteswar Rao Jerripothula, "SAMWave: Adapting Segment Anything Model to Difficult Tasks," in *Proceedings of British Machine Vision Conference (BMVC), 2025* **(Oral)**

Avi Gupta, Koteswar Rao Jerripothula and Tammam Tillo, "CIRCOD: Co-saliency Inspired Referring Camouflaged Object Discovery," in Proceedings of the *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025.*

Shyama Sastha, **Avi Gupta**, and Pankaj Jalote. "An Experience Report on Teaching a Large Introductory Programming Course in Hybrid Mode." In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 2, pp. 1704-1705. 2024.*

Under-Review

Toward Genuine Forgetting: Selective Unlearning in Large Language Models **2026**
Understanding how to unlearn requested knowledge in LLMs in a continual setting.

Generalizing Multimodal Foundational models for Class-Incremental Learning **2026**
Generalizing different foundational multimodal models in a context-driven class-incremental setting.

Camouflaged Object Detection *via* Multispectral Imaging **2026**
Introduce a novel approach that detects camouflaged objects from the corresponding multispectral inputs and predict corresponding binary mask.

SAMwave: Wavelet-Driven Feature Enrichment for Effective Adaptation of Segment Anything Model **IJCV**
Proposed real and complex-wavelet transformation for SAM, SAM2 and SAM3 on low-level vision tasks. **2026**
Saurabh Yadav[1], **Avi Gupta**[1] and Koteswar Rao Jerripothula

Reviewer

- NeurIPS
- CVPR
- ECCV
- TMLR
- TCSVT
- BMVC
- ICASSP
- ICPR
- CVPRW
- ICMLW

Achievements

Recognised among India's Top 100 AI/ML Researchers *(Identified by Amazon)* **2026**
Travel grant *(IndoML IIT Kharagpur)* **2026**
Doctoral Consortium *(BMVC 2026)* **2026**
DEI Grant *(BMVC 2026)* **2026**
Anveshan Setu Fellowship **2025**
Travel grant *(IndoML BITS Goa)* **2024**
GATE *(Qualified)* **2021, 2022**

References

Dr Tammam Tillo (TYUST, China) — *PhD Supervisor*
Dr Koteswar Rao Jerripothula (IIT, Kanpur) — *PhD Supervisor*
Dr Gautam Shroff (IIIT, Delhi) — *Academic Mentor*

[1]Equal Contribution